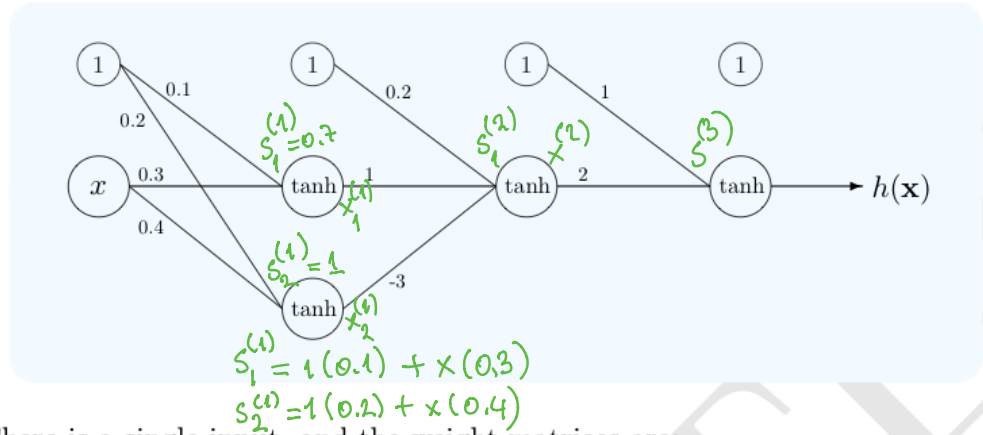
Example 7.1. Consider the following neural network.



There is a single input, and the weight matrices are:

$$\mathrm{W}^{(1)} = \begin{bmatrix} 0.1 & 0.2 \\ 0.3 & 0.4 \end{bmatrix}; \quad \mathrm{W}^{(2)} = \begin{bmatrix} 0.2 \\ 1 \\ -3 \end{bmatrix}; \quad \mathrm{W}^{(3)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

For the data point $x = 2, y = 1$, forward propagation gives:

$\mathbf{x}^{(0)}$	$\mathbf{s}^{(1)}$	$\mathbf{x}^{(1)}$	$\mathbf{s}^{(2)}$	$\mathbf{x}^{(2)}$	$\mathbf{s}^{(3)}$	$\mathbf{x}^{(3)}$
$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 0.1 & 0.3 \\ 0.2 & 0.4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.7 \\ 1 \end{bmatrix}$	$\begin{bmatrix} 1 \\ 0.60 \\ 0.76 \end{bmatrix}$	$[-1.48]$	$\begin{bmatrix} 1 \\ -0.90 \end{bmatrix}$	$[-0.8]$	-0.66

We show above how $\mathbf{s}^{(1)} = (\mathrm{W}^{(1)})^{\mathrm{T}}\mathbf{x}^{(0)}$ is computed. Backpropagation gives

$\boldsymbol{\delta}^{(3)}$	$\boldsymbol{\delta}^{(2)}$	$\boldsymbol{\delta}^{(1)}$
$[-1.855]$	$[(1 - 0.9^2) \cdot 2 \cdot -1.855] = [-0.69]$	$\begin{bmatrix} -0.44 \\ 0.88 \end{bmatrix}$

We have explicitly shown how $\boldsymbol{\delta}^{(2)}$ is obtained from $\boldsymbol{\delta}^{(3)}$. It is now a simple matter to combine the output vectors $\mathbf{x}^{(\ell)}$ with the sensitivity vectors $\boldsymbol{\delta}^{(\ell)}$ using (7.4) to obtain the partial derivatives that are needed for the gradient:

$$\frac{\partial \mathrm{e}}{\partial \mathrm{W}^{(1)}} = \mathbf{x}^{(0)}(\boldsymbol{\delta}^{(1)})^{\mathrm{T}} = \begin{bmatrix} -0.44 & 0.88 \\ -0.88 & 1.75 \end{bmatrix}; \quad \frac{\partial \mathrm{e}}{\partial \mathrm{W}^{(2)}} = \begin{bmatrix} -0.69 \\ -0.42 \\ -0.53 \end{bmatrix}; \quad \frac{\partial \mathrm{e}}{\partial \mathrm{W}^{(3)}} = \begin{bmatrix} -1.85 \\ 1.67 \end{bmatrix}.$$

Formulae for backpropagation:

$$\delta_1^{(L)} = 2(x_1^{(L)} - y_n)\left(1 - \theta^2(s_1^{(L)})\right)$$

$$\delta_i^{(\ell-1)} = \left(1 - \left(x_i^{(\ell-1)}\right)^2\right) \sum_{j=1}^{d^{(\ell)}} w_{ij}^{(\ell)} \delta_j^{(\ell)}$$

update weights: $\partial e / \partial w_{ij}^{(\ell)}$

$$w_{ij}^{(\ell)} \leftarrow w_{ij}^{(\ell)} - \eta \underbrace{x_i^{(\ell-1)} \delta_j^{(\ell)}}$$

$$= \frac{\partial e(w)}{\partial w_{ij}^{(\ell)}} = \frac{\partial e(w)}{\partial s_j^{(\ell)}} \times \frac{\partial s_j^{(\ell)}}{\partial w_{ij}^{(\ell)}} = \delta_j^{(\ell)} x_i^{(\ell-1)}$$